

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B.Tech. Theory (IT) Examination

December 2014

MA206 Probability, Statistics & Numerical Analysis

Date: 04/12/2014, Thursday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks:70

Instructions:

1. This question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever if required.
4. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Wright whether following statements are true or false. [07]

- (a) The sum of the deviations of 4 values of variable from 3 is zero therefore sum of four values is 4.
- (b) The number of ways three-lettered words could be made out of an alphabet of twenty-six letters, not using any letter more than once is $26 \times 26 \times 26$
- (c) If A and B are two events on sample space S with probability function $P(\cdot)$, then $P(A \cap B') = P(A) + P(B) - P(A \cap B)$.
- (d) If a random variable X has Poisson probability distribution with mean =2, the variance of distribution is 4.
- (e) The standard deviation of 10 values of variable is 12, if all values of variable are multiplied by 2, the new standard deviation of changed values of variable is 24.
- (f) When values of x are arranged in an increasing or decreasing order of magnitude, then middle-most value in this arrangement is called median of x
- (g) The value of Karl Pearson coefficient of correlation r satisfies $-1 \leq r \leq 1$.

Q.2. Answer the following questions. [14]

- (a) The following is the frequency distribution of the number of automobile accidents that occurred at 60 major intersections in a certain city during the Fourth of July weekend: Represent the frequency distribution by Histogram. [03]

Table 1: Frequency Distribution

Number of accidents	0	1	2	3	4	5	6	Total
Number of intersections	15	12	11	7	8	5	2	60

- (b) In a 2-week study of the productivity of workers, the following data were obtained on the total number of acceptable pieces which 100 workers produced: [04]

Table 2: Number of acceptable pieces

65	43	88	59	35	76	21	45	63	41
36	78	50	48	62	60	35	53	65	74
49	37	60	76	52	48	61	34	55	82
84	40	56	74	63	55	45	67	61	58
79	68	57	70	32	51	33	42	73	26
56	72	46	51	80	54	61	69	50	35
28	55	39	40	64	45	77	52	53	47
43	62	57	75	53	44	60	68	59	50
67	22	73	56	74	35	85	52	41	38
36	82	65	45	34	51	68	47	54	70

Group these figures into a distribution having the classes 20 – 29, 30 – 39, 40 – 49, ... 80 – 89.

- (c) The following data show amount (dollars) paid by users of commercial internet service in a month. [07]

Table 3: Monthly Expense \$ by 10 users

20	40	22	22	21	21	20	10	20	20
----	----	----	----	----	----	----	----	----	----

Determine following summary measures of monthly expense

- (i) Mean (ii) Median (iii) Mean deviation about 20 and Standard deviation.

OR

Q.2. Answer the following questions. [14]

- (a) Consider an example of a computer system with five identical tape drives. One possible random experiment consists of checking the system to see how many tape drives are currently available. Each tape drive is in one of two states: busy (labeled 0) and available (labeled 1). [03]

- (i) How many possible outcomes of this experiment?

- (ii) Suppose set E described by the statement "At least four tape drives are available", then write the members of E
- (b) The time it takes to transmit a file always depends on the file size. Suppose you transmitted 30 files, with the average size of 126 Kbytes and the standard deviation of 35 Kbytes. The average transmittance time was 0.04 seconds with the standard deviation of 0.01 seconds. The correlation coefficient between the time and the size was 0.86. Based on this data, fit a linear regression model and predict the time it will take to transmit a 400 Kbyte file. [04]
- (c) A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 10 students who took the regular course were recorded. [07]

Placement Test	50	35	35	40	55	65	35	60	90	35
Course Grade	53	41	61	56	68	36	11	70	79	59

- (i) Construct the scatter plot of the data.
- (ii) Compute Karl Pearson coefficient of correlation r using data in table.

Q.3. Answer the following questions. [14]

- (a) Consider a time-sharing computer system that allocates a finite quantum (or time slice) to a job scheduled for processor service, in an attempt to give fast service to requests for trivial processing. Observe $n = 8$ time-slice terminations, with success = "job has completed processing" and failure = "job still requires processing and joins the tail end of the ready queue of processes." Let p be the probability that "job has completed processing". If $p = 0.5$ then Write the formula to compute $P(X = r)$, r jobs have completed processing and compute the probability that [05]
- (i) no job has completer processing
- (ii) Exactly three jobs have completed processing
- (iii) atleast 2 jobs have completed processing
- (b) Assume that the number of messages input to a communication channel in an interval of duration t seconds is Poisson distributed with parameter $0.4t$. Compute the probabilities of the following events: [05]

- (i) Exactly three messages will arrive during a 10 second interval.
- (ii) At most 2 messages arrive in a period of 20 second.
- (c) An analog signal received at a detector (measured in microvolts) may be modeled as a Normal random variable $N(\mu = 200, \sigma = 16)$ at a fixed point in time. What is the probability that the signal will exceed 240 microvolts? You are given $P(Z \leq 2.5) = 0.99379$ [04]

OR

Q.3. Answer the following questions. [14]

- (a) Consider a normalized floating-point number in base (or radix) β so that the mantissa, X , satisfies the condition $\frac{1}{\beta} \leq X < 1$. Experience shows that X has the probability density: [07]

$$f(x) = \frac{k}{x}, k > 0$$

Determine

- (i) the value of k
- (ii) The probability $P(1 \leq X < \beta)$
- (b) Define Exponential distribution. Determine the mean and variance of exponential distribution. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 0.5$. What is the probability that a repair time does not exceed 2.5 hours? [07]

SECTION II

Q.4. Write whether following statements are true or false. [07]

- (a) If $f(x)$ is real and continuous in the interval (a, b) then length of interval in each iteration of Bisection algorithm is equal to $\frac{b-a}{2}$
- (b) The floating point representation of decimal number 0.625 in a hypothetical computer with precision $p = 3$, base $\beta = 2$ and $e_{max} = -1, e_{min} = 2$ is 1.01×2^{-1} .
- (c) If X is true value of quantity and $\hat{X}$ is its approximation, then relative error is defined as $X - \hat{X}$.
- (d) Let x_0, x_1, x_2, x_3 are equally spaced values and $f_k = f(x_k)$, the Simpson's 3/8 formula is $\int_{x_0}^{x_2} f(x)dx \approx \frac{h}{3}(f_0 + 3f_1 + 3f_2 + f_3)$
- (e) With usual notations one can write $\Delta y_0 = \nabla y_1$

- (f) Assume we have a table of values (x_i, y_i) , $i = 0, 1, 2$ of a function $y = f(x)$.

The Lagrange Polynomial is $L_2(x) = \sum_{i=0}^2 l_i(x)y_i$

Where $l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$

- (g) Assume we have a table of values (x_i, y_i) , $i = 0, 1, 2, 3$ of a function $y = f(x)$ and the values of x are equally spaced. By using these table values, we can determine Newton's forward difference interpolation polynomial with degree 3.

Q.5. Answer the following questions. [14]

- (a) Given the table of values [07]

x	2	2.5	3.0
$y = \ln x$	0.69315	0.91629	1.09861

evaluate $\ln 2.7$ using Lagrange's interpolation formula.

- (b) Find a root of the equation $xe^x - 1 = 0$ using Newton Raphson method. [05]
Perform four iterations starting with $x_0 = 1$
- (c) Define Absolute error, and relative error in numerical analysis. [02]

OR

Q.5. Answer the following questions. [14]

- (a) Fit a function of the form $y = ax^b$ to the following data. [07]

x	2	4	7	10	20	40	60	80
y	43	25	18	13	8	5	3	2

- (b) Find a real root of the equation $xe^x - 1 = 0$ using Bisection method, in the interval $[0,1]$. Perform five iterations. [05]
- (c) What is floating point representation of real number in digital computer? [02]

Q.6. Answer the following questions. [14]

- (a) From Taylor series for $y(x)$, find $y(0.1)$ correct to three decimal places if $y(x)$ satisfies [04]

$y' = \frac{dy}{dx} = x + y^2$ and $y(0) = 1$

- (b) Given $\frac{dy}{dx} = 1 + y^2$ where $y(0) = 0$, Find $y(0.2)$ and $y(0.4)$ correct to four decimals using the fourth order Runge-Kutta method. [05]

- (c) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal places using trapezoidal rule with $h = 0.25$ [05]

OR

Q.6. Answer the following questions. [14]

(a) Determine the value of y when $x = 0.1$ given that [04]

$$y(0) = 2 \text{ and } y' = \frac{dy}{dx} = y - x$$

by using modified Euler's formula. Take $h = 0.05$.

(b) Use four point Gaussian integration formula to approximate $\int_0^1 x dx$. [05]

Use following table values.

n	$\pm u_i$	W_i
4	0.339981	0.652145
	0.861136	0.347855

(c) Solve the equation $\frac{dy}{dx} = 1 + xy$, up to third approximation using Picard's [05]
method where $y(0) = 0$.